

CARGOMULE · DESIGN DECISIONS REGISTER

CargoMule design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cargomule		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried into the design: value engineering updated to USD 1,004; key switch and charge socket added to the things to confirm; one new open decision (gain selector and status light position) found while updating the appearance model
0.4	2026-10-02	Amish Chadha	Design for construction (CGM-DDR-003, Table 1, P1 to P15) accepted by Amish on 2026-10-02; added to decisions made
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for all eight open decisions on 2026-10-02; moved to decisions made; R8 fallback line in value engineering updated; 2026-09-26 review note items 1 to 3 closed
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the decision records,

the review note and the build plan
work

CargoMule design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

All open decisions in this register were decided on 2026-10-02. One new point was found on 2026-10-02 while carrying those decisions into the design:

#	Decision	Options	Recommendation	What it affects in the build	Source
9	Where the assist gain selector (G = 2 to 6) and the status light go. The 2026-09-26 renders put them on the enclosure's front face, which is now the door; the BOM puts them on the control board inside the locked box, where the rider can neither see the light nor change the gain without opening the door.	(a) leave them on the control board inside the box; (b) put them on the enclosure's right wall beside the key switch, with short leads inside the box like the key switch's; (c) a sealed selector and light on the deck's front edge, with a lead into the box through a gland.	(b): the rider reaches them at a stop with the key switch, no lead crosses the door's hinge, and two more panel holes cost little (about \$3). Proposed, awaiting Amish.	The enclosure's right wall (two more holes), the control board's leads, the appearance model and renders	Review note 2026-10-02

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The S-type load cell fits the 56.3 mm housing bore: body no more than 54 mm across its diagonal (about 51 x 13 x 61 mm assumed), M10 threads at both ends	The cell lives inside the coupler housing	CGM-DDR-003, P1
2	The hub motor: 100 mm spacing, 10 mm axle flats, disc mount, shell no more than 156 mm across within 16 mm of the disc-side locknut, cable exit on the non-disc side, torque arm in the kit	Sets the slot width, the caliper clearance and the cable route	CGM-DDR-003, P9, P10
3	The caliper's post-mount adapter for a 180 mm rotor: hole spacing on the dropout tab	The tab's two holes are drilled to suit	CGM-DDR-003, P9
4	The hitch's arm is 32 mm or less and takes the 6 mm locking pin; the hitch fits the bicycle's axle type	The arm goes 60 mm into the drawbar's 33 mm bore	CGM-DDR-003, P7; R7
5	The preload spring: about 30 N installed, free length about 96 mm, OD 40 mm or less, solid length under 40 mm	It sets the brake's onset and must not go solid within the 50 mm stroke	CGM-CAL-001, F3
6	Motor constants (winding, resistance, thermal) from the datasheet	They decide whether R3 holds (decided under CGM-DDR-002; part selection is TRL 4)	CGM-DDR-002, D12
7	Rivet nuts rated for 1.5 mm wall box section	They carry the deck, the enclosure and the lights	CGM-DDR-003, P8, P11
8	The key switch fits a 19 mm panel hole, and the 3-pin charge socket fits a 24 mm hole with its two M3 flange screws 24 mm apart and mates with the charger's plug; both sealed to IP65 with the cap closed	The two holes in the enclosure's right wall are cut to suit	CGM-DDR-003, A2; R11

■ Value engineering

Value-engineering target: USD 1,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 1,004 (USD 4 over the target).

Main cost drivers (CGM-CAL-001, I1): the battery pack (line 11, \$180), the hub motor wheel (line 7, \$160), the chassis frame (line 1, \$95), the deck and side boards (line 2, \$77) and the overrun brake coupler (line 6, \$58). The parts added to make the design buildable added \$18 (CGM-DDR-003), and the decisions of 2026-10-02 added \$7: the charge socket with its cap on the enclosure's right wall (about \$4) and the drawbar reflector (about \$3).

Savings worth trying:

- Salvaged 20 in wheels and brakes and a lighter enclosure, which the TRL 3 review estimated could bring the parts to about \$900.
- Straps or mesh in place of the plywood side boards (the trailer weighs 43.5 kg without the plywood side boards, CGM-DDR-003 A1 and CGM-CAL-001 A3, plus the straps or mesh), the fallback if the weighed prototype is over the 47 kg cap of R8 (CGM-DDR-003, A1).

- A SwapCell receiver in the fleet variant, which would price the shared pack once and leave it out of this kit (CGM-DDR-001, D2).

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D10: cost cuts then a \$1,000 budget, 36 V LiFePO4 384 Wh pack, steel frame, one hub motor in the left wheel, proportional assist G = 2 to 6, overrun mechanical brake plus assist cut, pitch reworded, drawbar force sensing, 20 in wheels with the hitch on the left axle end, handcart mode out of scope	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	CGM-DDR-001
2026-09-25	TRL 3 items D11 to D14: R8 relaxed to 45 kg with straps or mesh as a later weight option, thermal derating and a use limit for long climbs, 180 mm rotors with metallic pads, gain range kept at 2 to 6	Amish: "i accept all your recommendations, go with them across all repos."	CGM-DDR-002
2026-09-30	Make the design physically buildable while drawing the build plan; outstanding decisions go in this register, not the plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	CGM-DDR-003 (Draft; Table 3 and then Table 1 accepted on 2026-10-02, below)
2026-10-02	R8 set to 47 kg as a hard cap for the first prototype only, not a new target; the trailer is weighed at TRL 4, and straps or mesh side boards are the fallback if it weighs more than 47 kg	Amish: "i approve your recommendations for all 555 open decisions."	CGM-DDR-003, A1
2026-10-02	Key switch and charge port on the enclosure's right wall near the front, so no lead flexes across the door hinge	Amish: "i approve your recommendations for all 555 open decisions."	CGM-DDR-003, A2
2026-10-02	Coupler damping: friction washers on the lever pivot for the prototype; a rough-road chatter check is a pass condition at TRL 4, and the small hydraulic damper is fitted if the brake chatters or grabs	Amish: "i approve your recommendations for all 555 open decisions."	CGM-DDR-003, A3
2026-10-02	Appearance model and photoreal renders to be updated to the constructable design on Amish's Mac before the repo is made public	Amish: "i approve your recommendations for all 555 open decisions."	CGM-DDR-003, A4
2026-10-02	Keep the BOM's paint and sealed plywood for the prototype; the powder coat, faced boards, badges, tie-down tracks and gaiter are labeled as finished-product styling in the renders; the drawbar reflector is to be added to the BOM	Amish: "i approve your recommendations for all 555 open decisions."	Review note 2026-09-26, item 4
2026-10-02	Hero render composition accepted as it is; a kit close-up camera only if a load cell detail shot is wanted	Amish: "i approve your recommendations for all 555 open decisions."	Review note 2026-09-26, item 5
2026-10-02	First users chosen by one rule: a group moving 50 to 150 kg loads on short, repeated urban trips		CGM-DDR-001, O1

Date	Decision	Decided by	Record
	by bicycle or hand cart on paved, hilly routes. First candidate type to approach: a community food bank's local delivery or pantry runs; market traders are the second choice	Amish: "i approve your recommendations for all 555 open decisions."	
2026-10-02	Until a written legal check exists for the first partner's country, the trailer is for private ground only; the EU pedal-assist limits (250 W, assist only under pull, cut at 25 km/h, no throttle) stay the design rule as the strictest common case	Amish: "i approve your recommendations for all 555 open decisions."	CGM-DDR-001, O2
2026-10-02	Design for construction accepted: the changes P1 to P15 in Table 1, as made	Amish: "APPROVED: Design-for-construction changes in 10 repos (CityTwin, CoolShade, PalletPilot, Heliolite, PotholeLog, EarthPress, ReadyKit, CellCheck, CargoMule and ThermaCart)"	CGM-DDR-003, Table 1

The review note of 2026-09-26 also proposed a closed enclosure with a bottom or front access panel (item 1), controls on the enclosure's front face (item 2) and a clear guard over the load cell with an enclosure window (item 3). Items 1 and 3 are closed: CGM-DDR-003 overtook them (the enclosure has a front door, P8, and the load cell sits inside the coupler housing, P1). Item 2 is closed by the 2026-10-02 decision on CGM-DDR-003, A2.